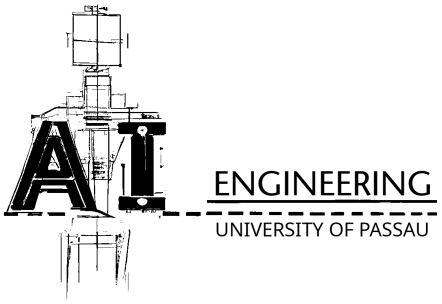
Mining Software Repositories

Exercise 5

Prof. Dr. Steffen Herbold
Winter 2026
Due on 18.11.2026



Author: Lukas Schulte

Task 1: The SZZ algorithm

The SZZ algorithm, named after the authors Sliverski, Zimmermann, and Zeller, is described starting on slide 39 of the 3rd chapter of the mining software repositories lecture. Further, the research paper first introducing the idea can be found here: `https://doi.org/10.1145/1083142.1083147`.

1.1 Implementing the SZZ algorithm

Implement the SZZ algorithm and use it to determine bug introducing commits for the project's issues. Since we have, at this point, not collected the changed lines in each commit, you may instead work on a file basis.

1.2 Theoretical exercise

1. Which assumptions does the SZZ algorithm make, and where do they break?
2. Take a look at the data in the appendix and apply the SZZ algorithm manually. Which bug introducing changes does it identify, and do breaking assumptions threaten the correctness of the result?

Task 2: Manual validation

Due to breaking assumptions, the results of the SZZ algorithm can be far from perfect. Before we can use the results for anything remotely important, we need to make sure that we reduce the noise in our data as much as possible.

1. Structure a reasonably sized subset of the results from task 1.1 in a human-readable format for manual validation. A simple solution will suffice here. Be prepared to explain how you would validate the results based on this.
2. Think about how you would present the results, if you would really have to validate hundreds of potential bug introducing commits.
3. Describe how manual validation can help to:
    a) Determine risk
    b) Mitigate risk

Appendix

Commits:

```
160f0625, 01.01.2023, author: lukas
"Initial commit for Project0815"
(1) + import version
(2) + VERSION = version(0.0001)

24f914fd, 02.01.2023, author: lukas
"ISSUE-0001: add option to perform multiplication"
(1) import version
(2) + import math
(2) VERSION = version(0.0001)
(3) + def multiply(a, b):
(4) +     return math.divide(a, b)

3f5b0912, 05.01.2023, author: alex
"ISSUE-0003: Change version to fix value 0.0002"
(1) - import version
(2) - import math
(3) - VERSION = version(0.0001)
(1) + import math
(2) + import version
(3) + VERSION = version(0.0002)
(4) def multiply(a, b):
(5)     return math.divide(a, b)

42d921a2, 06.01.2023, author: lukas
"Fixes ISSUE-0002: replace division by multiplication"
(1) - import math
(2) import version
(3) VERSION = version(0.0001)
(4) def multiply(a, b):
(5) -     return math.divide(a, b)
(5) +     return a * b

56682e28, 08.01.2023, author: lukas
"ISSUE-0004: bump to 1.0 for major release"
(1) import version
(2) - VERSION = version(0.0002)
(2) + VERSION = version(1.0)
(3) def multiply(a, b):
(4)   return a * b
```

Issues:

- ISSUE-0001: add option to perform multiplication
  created: 01.01.2023
  type: feature
  status: done
  assigned: lukas
- ISSUE-0002: multiplication actually performs division
  created: 03.01.2023
  type: bug
  status: fixed
  assigned: lukas
- ISSUE-0003: increase prototype version
  created: 04.01.2023
  type: feature
  status: done
  assigned: alex
- ISSUE-0004: bump version to 1.0 for bug-free release
  created: 07.01.2023
  type: feature
  status: done
  assigned: lukas
- ISSUE-0005: add option to perform division
  created: 08.01.2023
  type: feature
  status: open
  assigned: none

Timeline:

